INCH-POUND
A-A-59792A
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SUPERSEDING
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COMMERCIAL ITEM DESCRIPTION

TRUCK, LIFT, TELESCOPING MAST, CONTAINER HANDLER DIESEL-ENGINE-DRIVEN, PNEUMATIC-TIRED, 100,000-POUND CAPACITY

The General Services Administration has authorized the use of this Commercial Item Description (CID) for all federal agencies.

1. SCOPE. This CID) covers the general requirements for a commercial, diesel-engine-driven, telescoping mast, container-handling lift truck. The truck will primarily be used to lift, transport, and stack International Standards Organization (ISO) containers and sea/land containers of 20 and 40 foot lengths on paved and semi-improved surfaces. In standard configuration, the truck will be equipped with a top-handling attachment. In standard configuration, the truck will have the capability to lift, transport, and stack loads (not including attachment) up to 100,000 pounds at an 80-inch vertical load center.

2. SALIENT CHARACTERISTICS

2.1 Safety. The truck shall comply with UL 558 and OSHA standards in effect at the time of manufacture. An operator's seatbelt conforming to SAE J 386, and restraint devices designed to ensure the operator's upper body remains entirely within the protection of the overhead guard in the event of tip over shall be provided.

2.2 Transportability. The truck shall withstand, without damage highway; rail and marine transport.

2.3 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be in intimate contact with each other.

Beneficial comments, recommendations, additions, deletions, clarifications, etc., and any data which may improve this document should be sent to: U.S. AIR FORCE, AFLCMC/WN-SEV, 295 Byron Street, Robins AFB, GA 31098 or emailed to SPEC99@us.af.mil. Since contact information can change, you may want to verify the currency of this address using the ASSIST Online database at https://assist.dla.mil/ .

AMSC N/A FSC 3930

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2.4 Engine. The truck shall be powered by a standard, commercial, diesel engine. The engine shall operate on diesel fuel in accordance with ASTM D975 without detrimental effects on the engine, or its performance.

2.5 Fuel system. The truck shall have all necessary filters, water separators, and components required for diesel fuel operation. The truck shall have a fuel tank of sufficient capacity to allow at least eight hours of continuous operation.

2.6 Starting system. The starter switch shall not activate the engine starter while the engine is running, nor when the engine is not running and the transmission is in any forward or reverse gear. The starter switch shall only operate when the transmission is in the “neutral” or “park” positions.

2.7 Transmission. A torque converter and a power shift type of transmission shall be furnished, with at least three speeds forward and three speeds reverse. The transmission shall provide for positive inching control, or de-clutch control, that permits lifting of rated load at maximum lifting speed while the transmission control is in either forward or reverse direction, without vehicle movement. A transmission-oil cooler shall be furnished.

2.8 Hydraulic system. A pressure relief protection device is required along with pump(s), cylinders, control valves, filter(s), reservoir, hoses, and all other components necessary to make a complete hydraulic system. In the event of failure, or damage to the hydraulic system, a means to prevent the boom from automatically lowering or retracting and to prevent the load from tilting shall be provided. A manual override shall be provided to allow for load removal.

2.9 Visibility. Truck manufacture shall have a side-view of the truck showing sight lines for a 95th percentile male and a 5th percentile female. When requested, the manufacture should furnish a digital image of this drawing.

2.10 Boom attachment. Standard configuration of the truck shall include a Top-pick Attachment. It shall have at least 60 inches of lateral side shift ($\pm$ 30 inches to each side) and provide $\pm$ 5 degrees of rotation. The attachment shall use floating type universal twist locks and have a twist lock indicator mounted in the cab. The truck shall permit lifting only when all four twist locks are either locked or unlocked.

2.11 Steering. Power steering, with emergency steering in the event of power failure, shall be furnished.

2.12 Service brakes. The service braking system shall provide braking for all wheels. When operating in the forward and rearward direction, the system shall be capable of bringing the truck, with rated load in the carry position, to a complete stop within 21 feet under dry or wet conditions without any wheel leaving the ground and with fishtailing of no more than 1 foot from a speed of not less than 9 mph with a pedal force not to exceed 90 pounds.

2.13 Parking brake. Parking brake(s) shall hold, but not necessarily stop the truck without rated load when the truck is stopped on a 25 percent grade in both the ascending and descending

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position. The parking brakes shall be operated by controls in accordance with SAE J680 or shall be hand lever operated. The hand lever shall have a locking device. The manual control shall be located within reach of the operator and in a non-interfering position. The force required to pull the hand lever when setting the brake shall not exceed 68 pounds. The parking brake shall be a separate isolated system actuating the service brakes. Adjustment of parking brake shoes or bands shall be accomplished without the removal of any assembly.

2.14 Electrical system. A 24 volt direct current (VDC) electrical system shall be furnished. The electrical system shall consist of all electrical components necessary for operation of the truck. The truck shall have sufficient electrical grounding to prevent static discharge.

2.15 Backup alarm. An audible backup alarm shall be furnished. The alarm shall automatically activate when the transmission selector is placed in reverse.

2.16 Instruments and controls. In addition to the instruments supplied on the standard commercial truck, an hour meter shall be installed in the instrument panel. Except the hour meter, all instruments shall illuminate. All load motion controls shall be right hand operation controls and of the self-centering type, meaning; the controls shall return to the neutral position when released.

2.17 Lighting. Minimum of four front work lights, two headlights, and two rear work lights shall be furnished. An additional floodlight shall be furnished that is adjustable from inside the cab by the operator while seated; the floodlight shall provide at least 270° rotation capability. Tail light(s) and brake stop light(s) shall be installed. Individual operator controlled switches shall be provided for front floodlights, rear floodlights, and adjustable lights.

2.18 Horn. Horn button shall be mounted in the center of the steering wheel.

2.19 Operator's overhead guard. The guard shall be in accordance with ISO 3471. It shall be a rollover-protective-structure (ROPS).

2.20 Cab. The truck shall be equipped with an enclosed cab and shall have a door on each side of the vehicle with at least one window per door, which can be opened for ventilation. The doors shall be provided with hinges that can be serviced without door removal. In addition to the door windows, the cab shall be furnished with a front windshield, rear window, and side windows. The windshield and all windows shall be safety glass. The cab interior shall be furnished with water-resistant non-absorbent thermal insulation. A commercial fire extinguisher shall be mounted on the truck in an easily accessible location. The fire extinguisher shall be a minimum 2-1/2 pound capacity ABC type, or equivalent. The fire extinguisher shall be UL listed.

2.21 Windshield wipers. There shall be at least two wipers furnished, one on the front windshield, and one on the rear window.

2.22 Winterization kits. There shall be Type A (-65°F) and Type C (-25°F) options available for winter protection.

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2.22.1 Type A winterization. When specified (see 6.2), the truck shall be furnished with Type A winterization that shall protect to -65° F and shall consist of the following:

a. Power plant heaters. Engine coolant, engine oil and battery heaters shall be provided. All heaters shall operate on either (110 volts @ 60 Hz) or (220 volts @ 50 Hz) alternating current. A three wire, 25 feet long weatherproof cable of adequate capacity for all heaters being used simultaneously shall be provided. A stowage place shall be provided on the truck to keep the cable when not in use. The heaters shall be as follows:

1) The coolant heater shall be installed in the engine block or lower coolant inlet hose. A coolant-circulating pump, driven by a 110 volt or 220 volt alternating current motor, shall be provided when a coolant inlet hose heater is furnished. The heater shall have adequate capacity to maintain engine coolant at a temperature of +10° F in an ambient temperature of -65° F. It shall be controlled to limit engine coolant to not more than 150° F.

2) An engine oil heater with adequate capacity to maintain engine oil at a temperature of at least +10 degrees F in an ambient temperature of -65° F shall be furnished. It shall be controlled to limit engine oil temperature to not more than 150 ° F.

3) A battery heater shall be provided. It shall have adequate capacity to maintain battery electrolyte at a temperature of at least +10° F in an ambient temperature of -65°F, and shall have a thermostat to limit temperature of the electrolyte to not more than +80°F.

b. Cab. The cab shall meet the requirements of 2.20, and shall also contain thermal insulation, if necessary, to meet the heating requirements of 2.22.1.c.

c. Heater and defroster. The heater shall be of sufficient capacity to maintain a temperature of +40° F at cab floor level in an ambient temperature of -40° F.

d. Starting aid. Glow plug(s) or a measure shot ether injection system shall be furnished.

e. Antifreeze. Engine coolant shall be protected to -65° F.

2.22.2 Type C winterization. When specified (see 6.2), the truck shall be furnished with Type C winterization that shall protect to -25° F and shall consist of the following:

a. Cold starting aid. Glow plug(s) or a measure shot ether injection system shall be furnished.

b. Cab. The cab shall meet the requirements of 2.20, and shall also contain thermal insulation, if necessary, to meet the heating requirements of 2.22.2.c.

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c. Cab heater and defroster. The heater shall have sufficient capacity to maintain a temperature of at least +40° F at cab floor level in an ambient temperature of -20° F.

d. Antifreeze. Engine coolant shall be protected to -25° F with antifreeze.

2.23 Performance.

a. Capability to safely handle a 100,000-pound load, with an 80-inch load center, at all lifts heights.

b. Lifting speed: At least 30 fpm.

c. Lowering speed: Not more than 80 fpm with rated load on boom attachment.

d. Boom height (Horizontal): 195 inches maximum

e. Maximum boom height: 715 inches with rated load

f. Maximum boom lift height: 420 inches under spreader on first row

g. Right angle turn. With rated load on the boom, truck positioned perpendicular to a wall, and with the front of the load against the wall, the truck shall be able to back up and make a complete right angle turn (ending up parallel to the wall) within 345 inches.

h. Travel speed. Nine (9) miles per hour (mph) minimum in both forward and reverse, while carrying rated load on the boom. The truck speed shall be governed so that maximum speed for safe operations is not exceeded.

i. Slope ascension. With the truck facing up the slope and with rated load on the boom, the truck shall be able to accelerate up a 20 percent slope from a dead stop.

j. Ground clearance. Without a load on the boom, the truck ground clearance at the center of the wheelbase shall be at least 12 inches.

k. Load drift. With hydraulic fluid at normal operating temperature, the lift assembly shall hold rated load at maximum lift height and boom extension for at least 10 minutes with not more than 1.0 inch of cylinder travel, and not more than 0.5 degree of rotational drift.

l. Stability. With the truck stationary on a hard level surface, fueled to not more than one-quarter capacity and positioned with steering straight ahead, the rear wheels shall not clear the ground when: (1) 1.6 times rated load is suspended from the boom at a minimum 4 foot lift height or (2) rated load is tilted to maximum forward tilt from maximum rearward tilt at maximum lift height with engine at low idle speed for

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hydraulic mechanical drive and engine at operating speed for electric drive. Also the truck shall be capable of negotiating a 15 percent slope with and without rated load at maximum side shift position and oscillation centered in full circle operation at maximum steer angles in both directions without any wheel leaving the ground.

m. Noise limits. Maximum allowable noise level shall not exceed 84 dB (A) at the operator's ear. If the noise level at the operator's ear exceeds 84 dB (A) (85 to 92 dB (A)), a permanent warning decal or placard shall be installed inside the cab clearly visible to the operator. Noise level limits that exceed 92 dB (A) are not acceptable.

2.24 Painting. The truck shall be primed and painted using standard commercial practices. The topcoat shall be one of the following unless otherwise specified (see 6.2).

a. Forest Green (24052 of SAE AMS-STD-595)

b. Desert Sand (30313 of SAE AMS-STD-595)

c. Manufacturers standard commercial color

2.25 Rust proofing. When specified (see 6.2), tropical rust proofing shall be provided. This should be considered a premium rust proofing package usually consisting of an additional coating. The truck shall be treated with a corrosion preventative compound complying with A-A-59295. Some of these trucks may be used in tropical/coastal environments and experience high humidity and salt spray. These environments usually cause high levels of corrosion in the areas listed below. At a minimum, these areas shall be coated.

a. Cab, interior

b. Cab, exterior

c. Seams

d. Welds

e. Hidden recessed areas

f. Locations where dissimilar metals come in contact

g. Area above fuel tank(s)

h. Engine oil pan & transmission oil pan

i. Radiator support structure

j. Battery box

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2.26 Markings, data plates, and instruction plates. Corrosion resistant identification plate and all warning and instruction plates shall be provided and mounted on the truck. The identification plate shall contain the following information:

- Truck, Lift, Container Handler
- Capacity: (___) pounds at (___) inch load center
- Gross Vehicle Weight (GVW):
- National Stock Number (NSN):
- Contract Number:
- Serial Number:
- Model Number:
- Registration Number:
- Manufactured by:
- Name:
- Address:
- Service Phone Number:
- US Property:

2.27 Walkway coating. Floor plates and step surfaces shall be coated with a nonslip-coating compound, or be furnished with a nonslip metal or tread plate surface.

2.28 Slinging and tie down provisions. Permanently affixed slinging and tie down provisions that enable the truck to be lifted completely assembled and in its normal operational mode, and to be fastened to the floor or deck of a transportation medium, shall be provided. Provisions shall conform to MIL-STD-209, Classes 1 and 2 or Class 3 for Type II equipment.

2.29 Electromagnetic interference (EMI). The truck shall comply with EMI requirements of SAE J551-1 and SAE J551-2.

3. **REGULATORY REQUIREMENTS**

3.1 Recycled, recovered, or environmentally preferable, materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR). However, used, rebuilt or remanufactured components, pieces, and parts shall not be incorporated in the forklift.

3.2 Green Procurement Program. Green Procurement Program (GPP) is a mandatory federal acquisition program that focuses on the purchase and use of environmentally preferable products and services. GPP requirements apply to all acquisitions using appropriated funds, including services and new requirements. FAR 23.404(b) applies and states the GPP requires 100% of EPA designated product purchase that are included in the Comprehensive Procurement Guidelines list that contains recovered materials, unless the item cannot be acquired: a) competitively within a reasonable timeframe; b) meet appropriate performance standards, or c) at a reasonable price. The prime contractor is responsible for ensuring that all subcontractors comply with this requirement.

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4. PRODUCT CONFORMANCE PROVISIONS

4.1 Product Conformance. The products provided shall meet the salient characteristics of this CID, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial market. The unit will be in compliance with ISO 9001. The government reserves the right to require proof of such conformance.

5. PACKAGING

5.1 Preservation, Packaging, Labeling and Marking. Unless otherwise specified (see 6.2), the preservation, packaging, and packing shall be to a degree of protection to preclude damage to containers and/or content thereof under normal shipping conditions, handling, etc. This involves shipment from supply source to the receiving activity, and reshipment from the receiving activity. The preservation, packaging, and packing shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities, labeling and marking shall be as specified in the contract and/or order.

6. NOTES

6.1 Source of Documents.

6.1.1 SAE documents. SAE standards may be obtained at www.sae.org or from Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale PA 15096.

6.1.2 Federal Acquisition Regulation (FAR) documents. FAR information may be obtained at www.acqnet.gov/far or at the Superintendent of Documents, P.O. Box 371954, Pittsburgh, PA 15250.

6.1.3 Department of Defense and Federal documents. Department of Defense and Federal documents may be obtained from the DLA Document Services, Bldg. 4D, 700 Robbins Avenue, Philadelphia PA 19111-5094, or online at http://quicksearch.dla.mil .

6.1.4 OSHA Standards Occupational Safety and Health Association (OSHA). OSHA standards can be obtained at the U. S. Department of Labor, 200 Avenue NW, Room 423, Washington DC 20210), or online at http://www.osha.gov

6.1.5 Underwriters Laboratories (UL) Industrial Trucks, Internal-Combustion Engine Powered (Applications for copies should be sent to the Underwriters Laboratories INC., 333 Pfingsten Road Northbrook, IL 60062), or online at http://www.ul.com

6.1.6 ASTM standards can be obtained from ASTM, 100 Barr Harbor Drive, P O Box C700, West Conshohocken, PA 19428 or online at http://www.astm.org/ .

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6.1.7 Occupational Safety and Health Association (OSHA) standards can be obtained from OSHA, 200 Constitution Ave., Washington, D. C. 20210 or online at http://www.osha.gov/ .

6.2 Ordering data. Acquisition documents must specify the following:

a. Title, number, and date of this CID.

b. Type of Boom Attachment required if other than standard (see 2.10).

c. Winterization (type A or Type C) if required (see 2.22)

d. Fork Lift primed and painted, if different (see 2.24).

e. Rust proofing, if required (see 2.25).

f. Packaging requirements, if different (see 5.1).

6.3 Subject term (key word) listing.

Contracting mast
Load stacker, 100K capability
Solid, inflatable-tire
Truck, top-handling attachment

6.4 Changes from previous issue. The margins of this CID are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue

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